

ECON3200 Tutorial 7 Handout

Money in the Short Run: A New Keynesian Model

Worked solution for Tutorial 7, Exercise 7.1

Teresa Chang

Recap

The lecture shows that the New Keynesian model can be summarized by three equations:

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3)$$

These are, respectively:

- the **New Keynesian IS curve**,
- the **New Keynesian Phillips curve**,
- the **Taylor rule**.

The tutorial question

We are given the model

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t, \quad (3.NK^*)$$

with

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa}.$$

The economy starts at steady state in period 0. There is an **unanticipated negative monetary policy shock** in period 1, no productivity shocks, and the shock dies off by period 2. We are asked to compute

$$\hat{i}_1, \quad \hat{x}_1, \quad \hat{\pi}_1.$$

Step 1: Substitute in the parameter values

The tutorial gives:

$$\beta = 0.75, \quad \sigma = 1, \quad \phi = 1, \quad \delta_\pi = 2, \quad \delta_x = 0.5, \quad \rho_v = 0, \quad \kappa = 0.5.$$

First compute $\tilde{\kappa}$:

$$\begin{aligned}\tilde{\kappa} &= (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa} \\ &= (1 + 1) \frac{(1 - 0.5)(1 - 0.75 \cdot 0.5)}{0.5} \\ &= 2 \cdot \frac{(0.5)(0.625)}{0.5} \\ &= 2 \cdot 0.625 \\ &= 1.25.\end{aligned}$$

So the three equations become

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \hat{i}_t + E_t[\hat{\pi}_{t+1}], \quad (4)$$

$$\hat{\pi}_t = 0.75E_t[\hat{\pi}_{t+1}] + 1.25\hat{x}_t, \quad (5)$$

$$\hat{i}_t = 2\hat{\pi}_t + 0.5\hat{x}_t + v_t. \quad (6)$$

Lecture Reference

The lecture emphasizes that the whole New Keynesian economy can be reduced to **three equations**: an IS curve, a Phillips curve, and a Taylor rule. The first thing we do in the tutorial is therefore to plug in the parameter values and simplify that three-equation system.

Step 2: Use the fact that there are no productivity shocks

The tutorial states that there are **no productivity shocks**, so

$$\hat{z}_t = 0 \quad \text{for all } t.$$

From the lecture,

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t.$$

Hence

$$\hat{y}_t^f = 0 \quad \text{for all } t.$$

Also,

$$u_t = E_t[\hat{y}_{t+1}^f] - \hat{y}_t^f = 0.$$

Therefore the IS curve contains no u_t term in this question.

Lecture Reference

In the lecture, u_t captures movements in **flexible-price output**. If productivity does not move, then flexible-price output does not move either, so $u_t = 0$. This is why the demand side becomes much simpler in the tutorial question.

Step 3: Work out the monetary policy shock

The monetary policy shock process is

$$v_t = \rho_v v_{t-1} + \varepsilon_t.$$

Since $\rho_v = 0$, this becomes

$$v_t = \varepsilon_t.$$

The question says there is an **unanticipated negative monetary policy shock** at period 1, so

$$\varepsilon_1 = -1.$$

Therefore,

$$v_1 = -1, \quad v_2 = 0.$$

Lecture Reference

The lecture explains that v_t is the term in the Taylor rule that captures a **monetary policy shock**. A negative shock means the central bank sets a **lower nominal interest rate** than usual.

Note

There is a sign inconsistency in the tutorial slides: one line writes $v_1 = \varepsilon_1 = 1$, but the final worked solution uses $\varepsilon_1 = -1$, which is consistent with the question statement and the final answers. So in this handout we follow the final worked solution and use

$$v_1 = -1.$$

Step 4: Use the terminal condition

The tutorial says the shock dies off at period 2, and by then the economy is back at steady state. Hence

$$\hat{x}_2 = \hat{\pi}_2 = \hat{i}_2 = 0.$$

Therefore, in period 1,

$$E_1[\hat{x}_2] = 0, \quad E_1[\hat{\pi}_2] = 0.$$

Lecture Reference

The lecture's worked example solves the model **backwards**: once the terminal period is known, expectations of future variables become known numbers. That is exactly the logic used here.

Step 5: Solve for $\hat{x}_1$ using the NK IS curve

Start from the IS curve in period 1:

$$\hat{x}_1 = E_1[\hat{x}_2] - \hat{i}_1 + E_1[\hat{\pi}_2].$$

Since

$$E_1[\hat{x}_2] = 0, \quad E_1[\hat{\pi}_2] = 0,$$

we get

$$\hat{x}_1 = -\hat{i}_1.$$

Lecture Reference

The lecture interprets the NK IS curve as the **demand side** of the model. A higher interest rate makes households postpone consumption, so demand falls and the output gap falls. That is why the interest-rate term enters with a negative sign.

Step 6: Solve for $\hat{\pi}_1$ using the NK Phillips curve

Now use the Phillips curve in period 1:

$$\hat{\pi}_1 = 0.75E_1[\hat{\pi}_2] + 1.25\hat{x}_1.$$

Since $E_1[\hat{\pi}_2] = 0$, this becomes

$$\hat{\pi}_1 = 1.25\hat{x}_1.$$

Using $\hat{x}_1 = -\hat{i}_1$,

$$\hat{\pi}_1 = -1.25\hat{i}_1.$$

Lecture Reference

The lecture explains that the NK Phillips curve is **forward-looking**. Current inflation depends on expected future inflation and the output gap. Here expected future inflation is zero, so inflation is driven entirely by the current output gap.

Step 7: Plug into the Taylor rule

Now substitute everything into the Taylor rule:

$$\hat{i}_1 = 2\hat{\pi}_1 + 0.5\hat{x}_1 + v_1.$$

Using

$$\hat{\pi}_1 = -1.25\hat{i}_1, \quad \hat{x}_1 = -\hat{i}_1, \quad v_1 = -1,$$

we get

$$\begin{aligned}\hat{i}_1 &= 2(-1.25\hat{i}_1) + 0.5(-\hat{i}_1) - 1 \\ &= -2.5\hat{i}_1 - 0.5\hat{i}_1 - 1 \\ &= -3\hat{i}_1 - 1.\end{aligned}$$

Bring the interest-rate terms together:

$$4\hat{i}_1 = -1.$$

Hence

$$\hat{i}_1 = -0.25.$$

Lecture Reference

The Taylor rule represents **monetary policy**. Once the IS curve and Phillips curve have linked output and inflation to the interest rate, the Taylor rule closes the model and allows us to solve for equilibrium.

Step 8: Recover $\hat{x}_1$ and $\hat{\pi}_1$

From Step 5,

$$\hat{x}_1 = -\hat{i}_1 = -(-0.25) = 0.25.$$

From Step 6,

$$\hat{\pi}_1 = 1.25\hat{x}_1 = 1.25(0.25) = 0.3125.$$

Final Answer

$$\hat{i}_1 = -0.25, \quad \hat{x}_1 = 0.25, \quad \hat{\pi}_1 = 0.3125.$$

Step 9: Economic interpretation

The final signs make economic sense.

A **negative monetary policy shock** lowers the nominal interest rate:

$$\hat{i}_1 < 0.$$

Because prices are sticky in the short run, the real interest rate also falls. A lower real interest rate encourages households to bring spending forward, so demand rises. This creates a **positive output gap**:

$$\hat{x}_1 > 0.$$

Once output is above flexible-price output, firms face higher marginal cost, which pushes inflation up:

$$\hat{\pi}_1 > 0.$$

Lecture Reference

The lecture emphasizes the **interest-rate channel** of monetary policy:

policy rate $\rightarrow$ real interest rate $\rightarrow$ demand $\rightarrow$ output gap $\rightarrow$ inflation.

So the sign pattern in the tutorial is exactly what the lecture predicts:

$$\hat{i}_1 < 0, \quad \hat{x}_1 > 0, \quad \hat{\pi}_1 > 0.$$

Summary

In Tutorial 7.1, we solve the New Keynesian model by substituting in the parameter values, using the fact that there are no productivity shocks so $u_t = 0$, and then applying the terminal condition that the economy is back at steady state in period 2. This implies

$$E_1[\hat{x}_2] = E_1[\hat{\pi}_2] = 0.$$

The NK IS curve gives

$$\hat{x}_1 = -\hat{i}_1,$$

the NK Phillips curve gives

$$\hat{\pi}_1 = 1.25\hat{x}_1,$$

and the Taylor rule then implies

$$\hat{i}_1 = -0.25.$$

Therefore,

$$\hat{x}_1 = 0.25, \quad \hat{\pi}_1 = 0.3125.$$

Economically, a negative monetary policy shock lowers the interest rate, stimulates demand, creates a positive output gap, and raises inflation.